ASSIGNMENT 5

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[1]:
```
# Import required libraries
import os
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

1. Load the Titanic Dataset

[2]:
```
# Load the Titanic dataset directly from a public CSV link
# This avoids the need to install kagglehub.
import pandas as pd

url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.
  ↪csv"
df = pd.read_csv(url)

print("First five rows:")
display(df.head())
print("Dataset shape:", df.shape)
```

```
First five rows:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th…	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

```
Dataset shape: (891, 12)
```

2. Check Missing Values and Summary Statistics

[3]:
```
print("Missing values before cleaning:")
display(df.isnull().sum())

print("Numerical summary:")
display(df.describe())
```

```
Missing values before cleaning:
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
Numerical summary:
```

```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
```

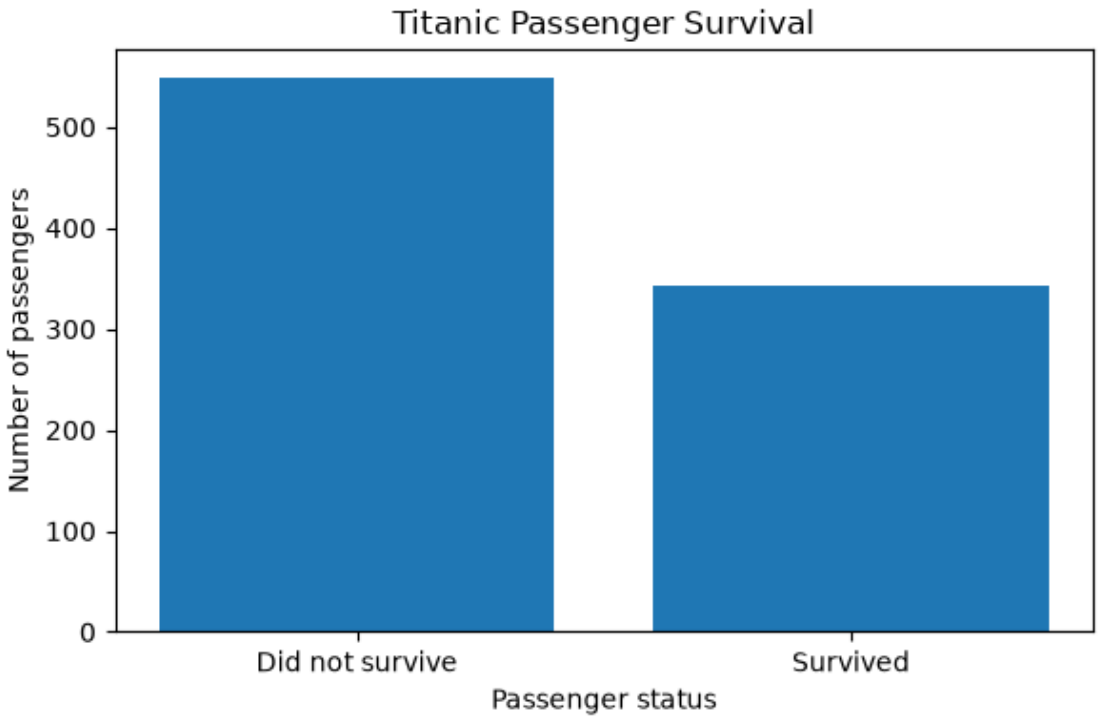
```
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

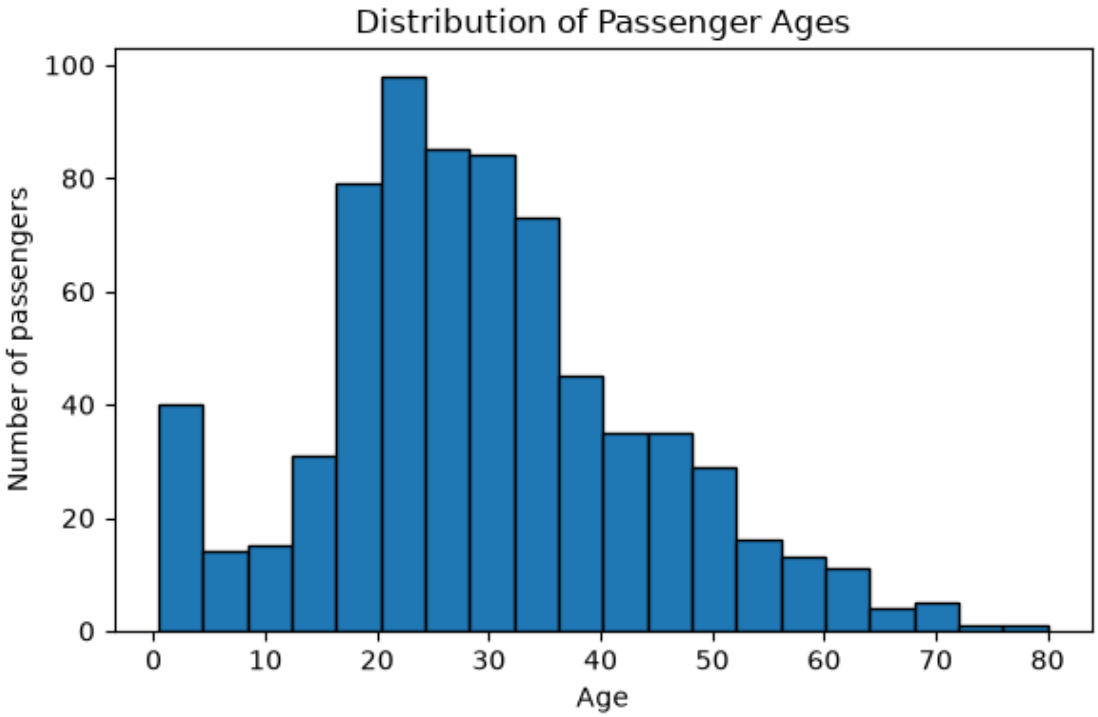
3. Explore the Data

[4]:
```
# Count passengers who survived and did not survive
survival_counts = df["Survived"].value_counts().sort_index()

plt.figure(figsize=(6, 4))
plt.bar(["Did not survive", "Survived"], survival_counts.values)
plt.title("Titanic Passenger Survival")
plt.xlabel("Passenger status")
plt.ylabel("Number of passengers")
plt.tight_layout()
plt.show()

# Distribution of passenger ages
plt.figure(figsize=(6, 4))
plt.hist(df["Age"].dropna(), bins=20, edgecolor="black")
plt.title("Distribution of Passenger Ages")
plt.xlabel("Age")
plt.ylabel("Number of passengers")
plt.tight_layout()
plt.show()
```





4. Handle Missing Values and Remove Unnecessary Columns

[5]:
```
# Fill missing numerical values with the median
df["Age"] = df["Age"].fillna(df["Age"].median())

# Fill missing categorical values with the most common value
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

# Remove columns that are not needed for this practical
df = df.drop(columns=["Cabin", "Name", "Ticket", "PassengerId"], errors="ignore")

print("Missing values after cleaning:")
display(df.isnull().sum())
print("Shape after cleaning:", df.shape)
```

```
Missing values after cleaning:

Survived    0
Pclass      0
Sex         0
Age         0
SibSp       0
```

```
Parch       0
Fare        0
Embarked    0
dtype: int64

Shape after cleaning: (891, 8)
```

5. Encode Categorical Columns

[6]:
```
# Convert text categories into numbers
sex_encoder = LabelEncoder()
embarked_encoder = LabelEncoder()

df["Sex"] = sex_encoder.fit_transform(df["Sex"].astype(str))
df["Embarked"] = embarked_encoder.fit_transform(df["Embarked"].astype(str))

print("Categorical columns encoded.")
display(df.head())
```

```
Categorical columns encoded.
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	1	22.0	1	0	7.2500	2
1	1	1	0	38.0	1	0	71.2833	0
2	1	3	0	26.0	0	0	7.9250	2
3	1	1	0	35.0	1	0	53.1000	2
4	0	3	1	35.0	0	0	8.0500	2

6. Separate Features and Target, Then Split the Data

[7]:
```
X = df.drop(columns=["Survived"])
y = df["Survived"]

# Split before scaling to avoid data leakage
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training feature shape:", X_train.shape)
print("Testing feature shape:", X_test.shape)
print("Training target shape:", y_train.shape)
print("Testing target shape:", y_test.shape)
```

```
Training feature shape: (712, 7)
Testing feature shape: (179, 7)
Training target shape: (712,)
Testing target shape: (179,)
```

7. Scale Numerical Features

[8]:
```
# Fit the scaler using training data only
scaler = StandardScaler()
scale_columns = ["Age", "Fare"]

X_train = X_train.copy()
X_test = X_test.copy()

X_train[scale_columns] = scaler.fit_transform(X_train[scale_columns])
X_test[scale_columns] = scaler.transform(X_test[scale_columns])

print("Age and Fare scaled successfully.")
display(X_train.head())
```

```
Age and Fare scaled successfully.
```

	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
331	1	1	1.253641	0	0	-0.078684	2
733	2	1	-0.477284	0	0	-0.377145	2
382	3	1	0.215086	0	0	-0.474867	2
704	3	1	-0.246494	1	0	-0.476230	2
813	3	0	-1.785093	4	2	-0.025249	2

8. Final Result

[9]:
```
print("Data preprocessing completed successfully.")
print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)
print("Missing values in training data:", int(X_train.isnull().sum().sum()))
print("Missing values in testing data:", int(X_test.isnull().sum().sum()))
```

```
Data preprocessing completed successfully.
Training data: (712, 7)
Testing data: (179, 7)
Missing values in training data: 0
Missing values in testing data: 0
```

Conclusion

The Titanic dataset was explored and preprocessed successfully. Missing values were handled, unnecessary columns were removed, categorical values were encoded, and numerical features were scaled. The dataset was split into training and testing sets before scaling to avoid data leakage.